

# A Bidirectional Neuro-Symbolic Foundation Pipeline for Continuous Sign Language AI

## Closed-Loop Spatial Scaffolding, Causal Memory, and Anatomy-Aware Diffusion

**Iniyan Joseph** · Applicant, Doctor of Science in Technology (D.Sc.)

Department of Computer Science, School of Science, Aalto University, Espoo, Finland

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**Full Application Dossier:** [Research Portal](#) | **Interactive 40-DOF Engine:** [WebGL Demo](#) | **LinkedIn:** [iniyandrews](#)

### Executive Abstract & Project Vision

Continuous Sign Language Processing—spanning forward Sign Language Production (SLP: Text-to-Sign) and backward Sign Language Translation (SLT: Sign-to-Text)—represents a fundamental frontier in multimodal artificial intelligence and assistive digital twin modeling. Current paradigms exhibit severe structural limitations: in production, systems oscillate between rigid avatars and unconstrained generative models (e.g., Pose-VQVAE, discrete diffusion) that suffer from quantization blur, spatial drift, and the smearing of centimeter-scale phonological minimal pairs; in translation, streaming translators suffer from **Spatial State Collapse**—purging historical 3D spatial reference loci ( $R$ -loci) under sliding windows—and **Environmental Grounding Blindness**. Crucially, current literature treats production and translation as disconnected, isolated engineering silos.

This doctoral project establishes a unified, closed-loop **Bidirectional Spatial Foundation Model (BiSign-LLM)** that unifies sign comprehension and synthesis into a single multimodal Transformer backbone. Rather than training a small model from scratch, we adapt a pre-trained open-weight frontier LLM (e.g., Llama 3.1 8B / Mistral NeMo 12B) via **Vocabulary Expansion** over an extended spatial-lexical vocabulary  $\mathcal{V}_{\text{extended}} = \mathcal{V}_{\text{base}} \cup \mathcal{V}_{\text{sign}}$ , inheriting extensive world commonsense and multilingual reasoning. The foundation model is anchored by two complementary modules: (1) an **Open-Domain Perceptual Tokenizer** crawling in-the-wild signing videos at scale (YouTube, Instagram, broadcast media) to parse continuous video into discrete phonological/spatial token sequences for web-scale continual pre-training on EuroHPC LUMI, coupled with a decoupled  $\mathcal{O}(1)$  **Causal Spatial Memory Bank** ( $M_t$ ) with deictic parity inversion ( $\mathbf{P}_{\text{deictic}}$ ) and *SceneGenie* 3D scene graphs; and (2) an **Embodied Biomechanical Actuator** compiling the LLM's predicted token streams into deterministic, continuous 3D articulation via closed-form 2-bone arm IK, the Kinematic Synthesizer (30 articulated joints) enforcing Battison's phonological laws (symmetry/dominance), a quartic Biomechanical Strain Objective ( $P_{\text{strain}}$ ), and manifold-projected residual diffusion ( $\mathbf{X}_{\text{final}} = \Pi_{\mathcal{M}}(\hat{\mathbf{P}} + \Delta\mathbf{X}_{\theta})$ ) to permanently eliminate non-physiological bone deformation and minimal-pair smearing. Both modalities execute natively inside the BiSign-LLM via standard causal language modeling. We outline a phased 4-year plan under Aalto's **3-Paper Cumulative Strategy** (ECCV → EMNLP → ICLR) leveraging CSC's **EuroHPC LUMI supercomputer** and participatory co-design with the Finnish Association of the Deaf (*Kuurojen Liitto ry*).

### Proof of Execution Ability: Interactive Kinematic Synthesizer · Live WebGL Demo: [Engine PoC](#)

To demonstrate immediate technical feasibility from Day 1 rather than presenting a speculative architecture, I have independently engineered and deployed an interactive Kinematic Synthesizer across 30 articulated phalangeal joints running client-side at 60 Hz in Three.js/WebGL. The engine validates the core neuro-symbolic premise: compiling symbolic sub-lexical sign descriptions into deterministic 3D hand configurations in  $\mathcal{O}(1)$  time ( $< 0.5$  ms) while enforcing Battison's phonological laws (bilateral symmetry and dominance constraints) and physiological joint limits. This working prototype confirms that the geometric foundations, 3D graphics rendering, and inverse kinematics solvers are already operational and de-risked.

## 1 Systemic Vision: An Assistive Communication Digital Twin

In critical societal environments—such as emergency clinical consultations, administrative hearings, and educational lectures—unimpeded cross-modal dialogue between Deaf signers and hearing professionals remains fundamentally constrained by the systemic shortage of qualified human interpreters. Consider an operational scenario in an emergency medical clinic: a triage clinician speaks aloud in Finnish. An assistive communication digital twin translates this speech into fluent, grammatically authentic Finnish Sign Language (*Suomalainen viittomakieli*, SVK) rendered via an interactive 3D digital human in real time ( $< 100$  ms latency). Conversely, when the Deaf patient responds in SVK, the system's vision channel tracks their articulated hand kinematics, facial grammar, and 3D spatial references, translating their discourse into spoken Finnish for medical staff.

My personal commitment to this problem stems from a conviction that multimodal AI must move beyond generic chat interfaces towards restorative physical communication—coupling continuous

biomechanics with foundation language models to restore communicative autonomy for an underserved linguistic community. Over **70 million Deaf and hard-of-hearing individuals** worldwide communicate primarily through natural sign languages [13, 45]. Spontaneous communication between Deaf signers and hearing individuals is virtually impossible without booking a professional interpreter days in advance. Under the World Health Organization (WHO) definition of health as a state of complete physical, mental, and social well-being, this persistent communication barrier creates dangerous medical vulnerabilities and profound social exclusion. In Finland and across Europe, equal communicative access is constitutionally and statutorily mandated under **Finland's Sign Language Act (Viittomakielilaki 359/2015)** and the European Accessibility Act (Directive 2019/882).

### 1.1 Fundamental Representational Bottlenecks in Sign Language AI

While spoken languages enjoy trillions of words digitized across books, Wikipedia, and the web, **sign languages possess no conventional written form**. They are 4D spatio-temporal languages operating simultaneously across physical space [13, 26, 45]: coordinating high-DOF articulated finger kinematics, subtle facial expressions (eyebrow raises AU1/AU2/AU4), and virtual 3D reference anchors in the air (*R*-loci) [36]. Prior machine learning paradigms suffer from two complementary structural failure modes:

1. **Generative Synthesis Artifacts (Non-Physiological Bone Deformation)**: When unconstrained generative models (diffusion or GANs) attempt to generate sign movements directly from text, they lack physiological and kinematic manifold constraints. Digits deform into physically impossible postures (such as phalangeal stretching and joint dislocations), smearing delicate centimeter-scale minimal pairs (e.g., inverting MOTHER at the chin into FATHER at the forehead).
2. **Perceptual Spatial State Collapse (Truncated Temporal Memory)**: Signers assign discourse referents to persistent 3D spatial coordinates in signing space. Contemporary video translation models process footage through short sliding windows that purge past frames every few seconds, causing the system to lose track of discourse referents and completely scrambling directional agreement verbs like INVITE or VISIT.

### 1.2 The Tripartite Foundation Architecture

Hosted at the **ELLIS Institute Finland** and the **Department of Computer Science at Aalto University** under the supervision of **Dr. Azade Farshad**, this 4-year doctoral project builds this complete, closed-loop conversational bridge. We solve the representation bottleneck by functionally decoupling the system into three unified foundation modules:

- ▶ **1. Articulatory Perception Channel (Perceptual Tokenizer)**: Watches continuous multi-channel signing video at 60 Hz, tracking articulated hand kinematics and facial blendshapes, while maintaining an external  $\mathcal{O}(1)$  Causal Spatial Memory Bank ( $M_t$ ) with deictic parity inversion ( $\mathbf{P}_{\text{deictic}}$ ) to preserve persistent 3D referents immune to temporal window truncation.
- ▶ **2. Multimodal Foundation Reasoning Core (BiSign-LLM)**: Rather than training an LLM from scratch without world priors, we adapt a frontier open-weight model (**Llama 3.1 8B / Mistral NeMo 12B**) via **Vocabulary Expansion** ( $\mathcal{V}_{\text{extended}} = \mathcal{V}_{\text{base}} \cup \mathcal{V}_{\text{sign}}$ ) on the EuroHPC LUMI supercomputer, enabling it to natively understand 3D spatial grammar and translate fluently between spoken text and sign discourse.
- ▶ **3. Embodied Biomechanical Actuator (Synthesis Channel)**: Compiles predicted sign macro-tokens into an anatomically constrained 3D digital human at 60 Hz on WebGL. Governed by Battison's phonological symmetry laws, analytical 2-bone arm IK with humeral swivel orbits, and physiological joint limit regularizers, it guarantees anatomical bone-length invariance and natural, authentic biological motion.

## 2 Linguistic Taxonomy & 3D Spatial Phonology

To satisfy the strict grammatical standards of native Deaf signers, computational representations must be grounded in formal sign language linguistics.

### 2.1 Sub-Lexical Parameter Composition & Minimal Pairs

In formal sign linguistics (Stokoe, 1960 [45]; Battison, 1978 [4]), every lexical sign is decomposed into five discrete phonological parameters:  $S = \langle \mathbf{H}, \mathbf{L}, \mathbf{M}, \mathbf{O}, \mathbf{F}_{\text{NMS}} \rangle$ , where **H** represents Handshape



(a) ASL FATHER (single tap)



(b) ASL GRANDFATHER (double-arc outward bounce)

Figure 1: **Phonological Minimal Pairs:** Identical handshape and forehead location require millimetric spatial precision ( $r_{\text{tol}} \leq 5 \text{ mm}$ ) and high-frequency temporal acceleration tracking to prevent semantic inversion between kinship relations.



Figure 2: **Non-Manual Signals (NMS) as Visual Prosody:** Multi-channel face tracking capturing sub-pixel eyebrow Action Units (AU1+AU2 vs. AU4 for conditionals vs. questions), 3D ocular gaze raycasting vectors ( $\mathbf{g}_{\text{gaze}}$ ) locking onto referential loci, and mouth morphemes.

(finger joint angles),  $\mathbf{L}$  denotes 3D anatomical Location target,  $\mathbf{M}$  is Movement trajectory primitive,  $\mathbf{O} \in \text{SO}(3)$  is wrist Orientation, and  $\mathbf{F}_{\text{NMS}} = \langle \mathbf{b}, \mathbf{m}, \mathbf{g}, \mathbf{h} \rangle$  encodes facial Action Units (brows  $\mathbf{b}$ , mouth  $\mathbf{m}$ , gaze  $\mathbf{g}$ , head tilt  $\mathbf{h}$ ).

Signs differing in exactly one parameter are **phonological minimal pairs**, where centimeter-scale deviations invert semantic meaning:

- ▶ **Location Minimal Pairs:** In American Sign Language (ASL), MOTHER is articulated with an open-5 handshape contacting the chin, whereas FATHER contacts the forehead. A vertical deviation of mere centimeters inverts maternal to paternal kinship.
- ▶ **Movement Minimal Pairs:** FATHER involves a single contact tap at the forehead, whereas GRANDFATHER shares the identical handshape and location but executes a double-arc outward ballistic bounce, requiring high-frequency temporal acceleration tracking.
- ▶ **Orientation Minimal Pairs:** NAME involves two H-handshapes tapping transversely; altering palm orientation from inward to downward produces an ungrammatical form.

## 2.2 Non-Manual Signals & 3D Spatial Syntax (R-loci)

Non-manual signals act as the visual equivalent of spoken prosody and syntactic clause delimiters [34,35]: raised eyebrows (AU1+AU2) mark Yes/No questions and conditionals, furrowed eyebrows (AU4) mark Wh-questions, and ocular gaze ( $\mathbf{g}_{\text{gaze}}$ ) raycasts toward empty spatial loci to establish pronominal reference before manual indexing [13,44].

As modeled by Pfau et al. (2018) [36] and Lillo-Martin & Meier (2011) [30], signers project

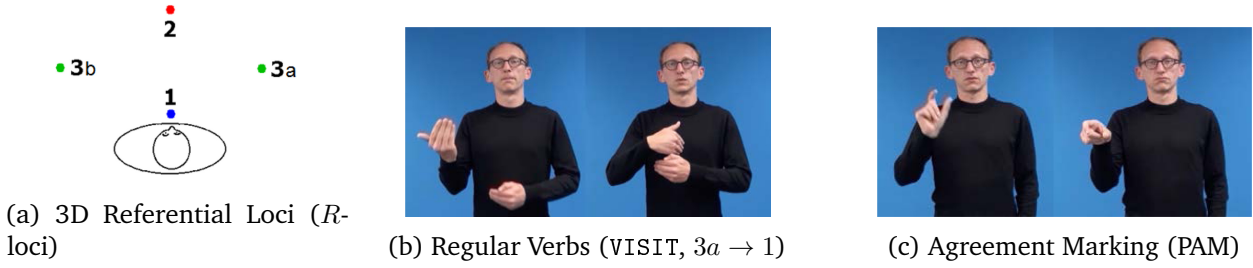


Figure 3: **Syntactic 3D Spatial Loci and Agreement Trajectories** [36]: (a) 3D coordinate anchors in the signing volume; (b) regular agreement verbs traversing from Subject to Object locus ( $3a \rightarrow 1$ ); (c) plain verbs utilizing auxiliary pointing markers (Person Agreement Marker, PAM) to project grammatical agreement.

coordinate anchors into the air around their body, termed **Referential Loci (*R*-loci)**. In our interaction volume:

- ▶ Locus<sub>1</sub> (Signer Sternum):  $[0.00, 0.00, 0.00]$  m, designating first-person (“I”, “Me”).
- ▶ Locus<sub>2</sub> (Addressee):  $[0.00, 0.60, 0.00]$  m, designating second-person (“You”).
- ▶ Locus<sub>3a</sub>, Locus<sub>3b</sub> (Third-Party Referents): Right hemispace  $[+0.35, 0.45, +0.05]$  m and Left hemispace  $[-0.35, 0.45, +0.05]$  m.

Directional verbs physically traverse these loci: **Regular verbs** (GIVE, VISIT) move from Subject to Object (Locus<sub>3a</sub> → Locus<sub>1</sub> for “Mother visits me”); **Backward verbs** (INVITE, TAKE) invert physical trajectory relative to agency, moving from Goal to Agent (Locus<sub>3a</sub> → Locus<sub>1</sub> for “I invite Mother”) [32, 36]; and **Orientation-only verbs** (LOOK-AT, WARN) aim palm or index vectors directly toward the active referent.

### 2.3 Empirical Regularities in Sign Lexicons (ASL-LEX 2.0)

To formalize sub-lexical decomposition, our parameter space is calibrated against **ASL-LEX 2.0** (2,723 signs across 54 dimensions [7, 41]). Prior linguistic analyses reveal critical structural constraints:

1. **Handshape Distribution Skew:** In accordance with Battison’s phonological laws [4], the top 6 unmarked handshapes (B, A/S, 1, 5, 0, C) represent  $\approx 52\%$  of the lexicon, and the top 12 cover  $\approx 78\%$ . Calibrating 12 anatomical finger configurations provides immediate coverage of conversational vocabulary.
2. **Spatial Location Clustering:** Signs cluster tightly:  $\approx 40\%$  occur in neutral space before the chest,  $\approx 34\%$  cluster around cranial/facial landmarks (chin, forehead, cheek), and the remainder occur on the trunk or base hand, motivating millimetric precision on cranial zones.
3. **Battison’s Symmetry Constraints:** Two-handed asymmetric signs restrict the passive base hand to unmarked resting shapes, dramatically simplifying multi-hand coordination.
4. **Movement Dynamics:** Approximately  $62\%$  of signs involve single continuous paths (verbs), while  $38\%$  feature repeated contact taps or oscillations (nouns), parameterized via a repetition multiplier.

## 3 Literature Review & Rigorous Scientific Gap Analysis

### 3.1 Generative Production (SLP): Procedural Scaffolding vs. Neural Diffusion

Generative SLP has oscillated between procedural correctness and perceptual realism. Early procedural animation systems (VisiCast, JASigning) converted phonetic scripts (HamNoSys/SiGML) into deterministic avatars [31]. While guaranteeing bone-length invariance and joint limits, they produced robotic motion lacking biological momentum and co-articulation. Recent neural approaches using autoregressive Transformers, GANs, or discrete latent diffusion (e.g., G2P-DDM [50], Neural Sign Actors [3]) mapped glosses to vector-quantized latent spaces (Pose-VQVAE). However, codebook quantization introduces spatial blur that smears millimeter-scale minimal pairs. Conversely, continuous unconstrained skeletal diffusion models (e.g., MDM [47]) diffuse in Euclidean space  $\mathbb{R}^{J \times 3}$ ; lacking kinematic manifold projection, they stretch bone lengths (“rubber fingers”) and breach physiological Range-of-Motion (ROM) limits. Simulator-guided diffusion (e.g., PhysDiff [52], BioMoDiffuse [5]) integrates rigid-body physics (Isaac Gym / OpenSim) for gross whole-body locomotion, but rigid-body simulators cannot resolve high-frequency 20-DOF finger dexterity, Kapandji thumb opposition, or delicate sub-centimeter ( $r_{\text{tol}} \leq 5$  mm) cranial contact boundaries where rigid contact penalties cause numerical chattering. Similarly, while

recent transition diffusion models like StgcDiff [23] apply spatial-temporal graph convolutions (ST-GCN) as soft neural conditioning, soft feature conditioning cannot mathematically guarantee hard bone-length conservation ( $\ell_{ij}$ ) or prevent anatomical dislocations under fast strokes. Appearance-level models (e.g., DiffSign [27]) render photorealistic 2D RGB pixels from 2D poses, bypassing 3D kinematic fidelity. Our solution is **Manifold-Projected Continuous Residual Diffusion** ( $\mathbf{X} = \Pi_{\mathcal{M}}(\hat{\mathbf{P}} + \Delta\mathbf{X}_{\theta})$ ), anchoring diffusion to exact procedural coordinates while enforcing iterative holonomic manifold constraints.

### 3.2 Perceptual Translation (SLT) & Multimodal Sign Foundation Models

Current SOTA Sign Language Translation treats the mapping from video to text as an end-to-end sequence-to-sequence problem ( $\mathbf{V}_{1:T} \rightarrow \mathbf{W}$ ). In conversational settings, existing models exhibit three fundamental structural failure modes:

- 1. Spatial State Collapse (The Streaming Bottleneck):** As empirically quantified by **Ranum, Hadfield, & Bowden (2026)** [38], real-time streaming translation ( $< 300$  ms latency) under sliding temporal window truncation purges established spatial loci, inducing up to **74% directional agreement failures** on referential verbs (see also Kermani et al. [25]). While recent prompting heuristics (e.g., SpaMo [43]) append static frame-level 2D/3D Euclidean distances to LLM prompts, static text prompting lacks a stateful, persistent 3D spatial register that survives across multi-turn discourse. We resolve this by introducing our decoupled  $\mathcal{O}(1)$  **Causal Spatial Memory Bank** ( $M_t$ ), which persistently tracks 3D referent coordinates outside the temporal attention window.
- 2. Environmental Grounding Blindness:** Pure-skeletal translators [12,51] are blind to external physical space. When a signer points to a room object or uses an ambiguous depicting classifier (e.g., CL:C, representing a cup, cylinder, or vase) [46], skeletons lack visual grounding to identify the referent.
- 3. Unidirectional Disconnection in Foundation Sign LLMs:** Recent multimodal foundation efforts bifurcate into disconnected paradigms: text-to-pose generation models (e.g., SignLLM [14]) generate raw 2D/3D skeletal keypoints directly from text prompts without 3D spatial referential loci, causing token explosion ( $> 3,000$  values/s) and phonetic minimal-pair smearing; conversely, recent sign-to-text translation (SLT) foundation models (e.g., ViPo-MLLM [48], SignBind-LLM [42]) align visual-pose features to language models, yet remain strictly unidirectional recognition systems lacking explicit 3D spatial memory ( $M_t$ ) or bidirectional closed-loop 3D kinematic synthesis.

| Research Challenge           | Current SOTA Paradigm                                        | Structural Limitation                                                                            |
|------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Quantization Blur            | Pose-VQVAE + Discrete Diffusion (G2P-DDM [50])               | Codebook quantization rounds continuous coordinates, smearing minimal pairs.                     |
| Unconstrained Drift          | Unconstrained Motion Diffusion (MDM [47])                    | Euclidean diffusion without kinematic manifold stretches bones (“rubber fingers”).               |
| Locomotion vs. Dexterity     | Simulator-Guided Diffusion (PhysDiff [52], BioMoDiffuse [5]) | Rigid simulators model gross locomotion, failing on 20-DOF dexterity and cranial contacts.       |
| Soft Graph Priors            | Graph-Conditioned Diffusion (StgcDiff [23])                  | Soft ST-GCN conditioning cannot mathematically guarantee bone-length invariance ( $\ell_{ij}$ ). |
| Spatial State Collapse       | Sliding-Window / Heuristics (Ranum [38], SpaMo [43])         | Truncating context windows or static prompting purges historical $R$ -loci across turns.         |
| Environmental Blindness      | Pure-Skeletal Translators (STMC [51])                        | Skeletons lack environmental context to resolve depicting classifiers (CL:C).                    |
| Unidirectional Disconnection | Text-to-Pose or SLT-only LLMs (SignLLM [14], ViPo-MLLM [48]) | Lack 3D loci memory ( $M_t$ ), 20-DOF kinematics, and bidirectional closed-loop synthesis.       |

Table 1: Core Algorithmic Challenges and SOTA Failure Modes in Sign Language AI.

## 4 Proposed Technical Methodology: The Unified Foundation Architecture

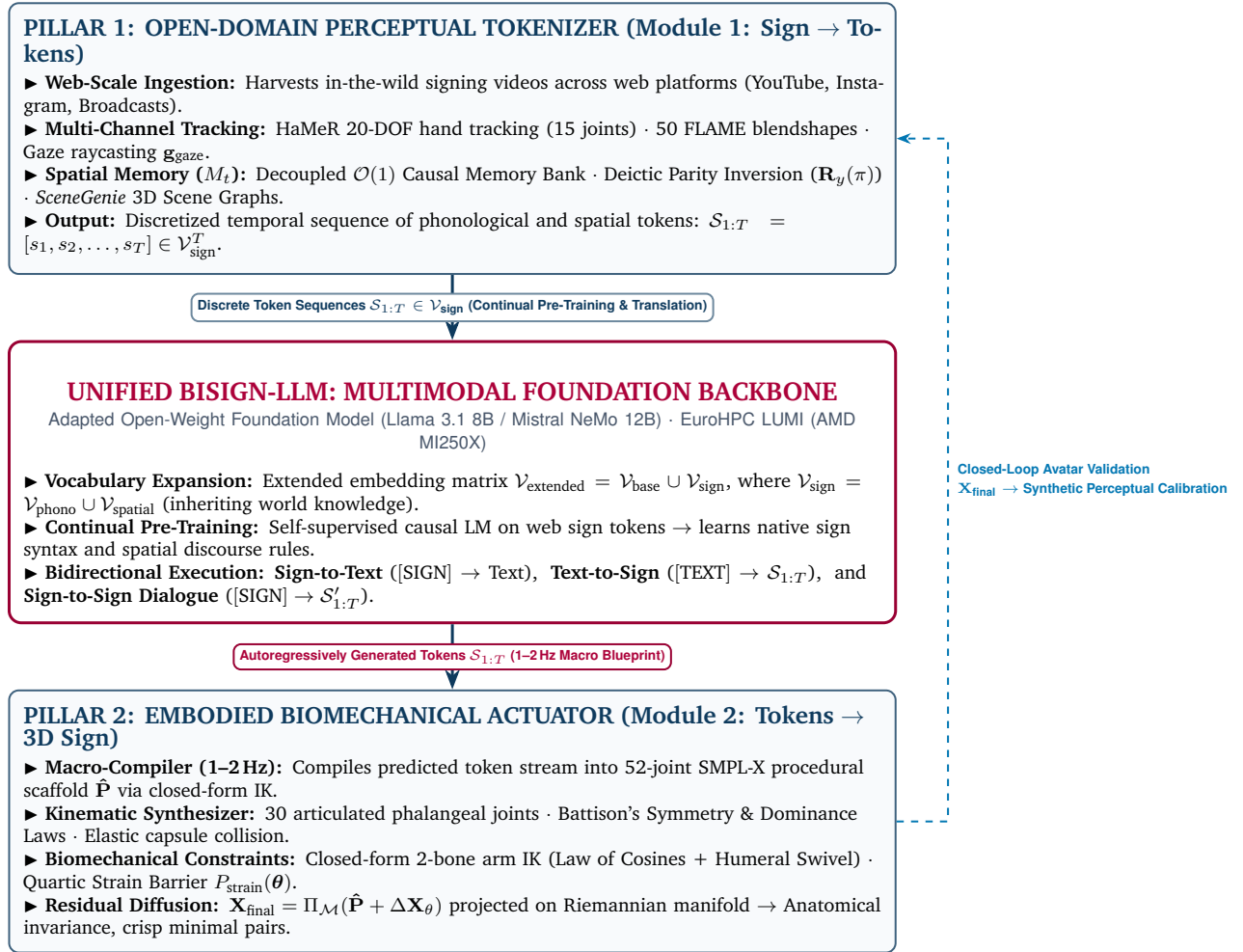


Figure 4: **The Unified BiSign Foundation Architecture:** Combining web-scale perceptual tokenization (Module 1), an adapted open-weight multimodal foundation model ( $\mathcal{V}_{\text{extended}}$ ), and a deterministic 40-DOF biomechanical actuator (Module 2) into a closed-loop bidirectional system.

#### 4.1 Architectural Blueprint: The Unified BiSign Foundation Loop

The overarching breakthrough of our architecture is that sign language is treated as a **first-class native tokenized modality within a unified foundation model**, rather than two disconnected engineering pipelines. As formalized in Figure 4, the architecture couples three seamlessly integrated pillars:

1. **Module 1: The Open-Domain Perceptual Tokenizer (Sign  $\rightarrow$  Tokens):** An automated perceptual front-end that harvests open-domain signing videos at web-scale (YouTube, Instagram Reels, broadcast media), recovering 20-DOF hand articulations (15 joints), 50 FLAME blendshapes, and continuous 3D loci raycasting into an external  $\mathcal{O}(1)$  Causal Spatial Memory Bank ( $M_t$ ). It converts raw, unlabelled video into discrete temporal sequences of phonological and spatial tokens  $\mathcal{S} \in \mathcal{V}_{\text{sign}}$ .
2. **The Unified BiSign-LLM Backbone:** An open-weight foundation model (adapted from Llama 3.1 8B or Mistral NeMo 12B) with an expanded embedding matrix  $\mathcal{V}_{\text{extended}} = \mathcal{V}_{\text{base}} \cup \mathcal{V}_{\text{sign}}$ . Continually pre-trained on billions of extracted web-scale sign tokens on EuroHPC LUMI, the model natively masters sign grammar, spatial syntax, and bidirectional translation (Sign-to-Text, Text-to-Sign, and direct Sign-to-Sign conversational dialogue).
3. **Module 2: The Embodied Biomechanical Actuator (Tokens  $\rightarrow$  3D Sign):** When the LLM autoregressively predicts sign tokens at a 1–2 Hz macro-rate, our procedural kinematic engine compiles them into continuous 60 Hz 3D animation via closed-form 2-bone arm IK, the Kinematic Synthesizer (30 articulated joints) respecting Battison’s phonological laws, a quartic Biomechanical Strain Objective ( $P_{\text{strain}}$ ), and manifold-projected residual diffusion ( $\mathbf{X}_{\text{final}} = \Pi_{\mathcal{M}}(\hat{\mathbf{P}} + \Delta \mathbf{X}_{\theta})$ )—permanently eliminating non-physiological bone deformation and spatial blur.

## 4.2 The Forward Pass: Text-to-Sign Language Production (SLP)

The Forward Pass transforms spoken text into a continuous 60 Hz 3D avatar stream across four specialized stages:

### 4.2.1 Stage 1: Linguistic Parsing, 3D Spatial Registry & Generative LLM OOV Fallback

Natural sign languages function as Topic-Prominent languages governed by **Time-Topic-Comment (TTC)** structure [19, 29, 34]. Feeding linear spoken Subject-Verb-Object (SVO) text directly to an avatar produces ungrammatical pidgin. Stage 1 executes three operations:

1. **Syntactic Transfer Grammar (SVO  $\rightarrow$  TTC):** Extracts temporal adverbials and fronts them to clause-initial position to establish the discourse timeline, deletes non-signing copulas (*is*, *are*, *was*), and shifts interrogative markers clause-final where facial markers (AU4) frame the proposition.
2. **Coreference Resolution & 3D Spatial Registry:** Allocates coordinate anchors in the signer's interaction volume:  $\text{Locus}_1 = [0.00, 0.00, 0.00]$  m (Self),  $\text{Locus}_{3a} = [+0.35, 0.45, +0.05]$  m (Primary referent), and  $\text{Locus}_{3b} = [-0.35, 0.45, +0.05]$  m (Secondary referent). Directional verbs bind trajectory end-points directly between these coordinates. The sentence is compiled into a 3D spatial scene graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$  encoded via a Relational Graph Transformer (RGT).
3. **Generative LLM Out-of-Vocabulary (OOV) Procedural Fallback:** Rather than freezing or falling back to tedious 8-second letter fingerspelling when encountering unseen terms [9, 33], we leverage the spatial commonsense reasoning of LLMs to synthesize **Depicting Signs (Classifier Predicates)** [13, 46] from atomic sub-lexical parameters. Crucially, the LLM does not emit continuous joint Euler angles directly; rather, it outputs high-level discrete phonological tokens—handshape primitives (**H**), movement paths (**M**), palm orientations (**O**), and contact locations (**L**)—which are subsequently validated against phonological collision rules and compiled into continuous kinematics by Stage 2.

Listing 1: Validated Pydantic 4D Phonetic JSON Interface with Procedural OOV Synthesis.

```
{
  "sentence_id": "AALTO_SVK_001", "asl_gloss": "YESTERDAY MOTHER 3a-VISIT-1",
  "sentence_nms": {"eyebrows": "NEUTRAL", "negation": false},
  "spatial_registry": {
    "Locus_3a": {"entity": "MOTHER", "coord": [0.35, 0.45, 0.05]},
    "Locus_1": {"entity": "SELF", "coord": [0.00, 0.00, 0.00]}
  },
  "tokens": [
    {"lemma": "YESTERDAY", "handshape": "HS_A_THUMB", "loc": [0.15, 0.10, 0.30], "mov": "CHEEK_TAP"},
    {"lemma": "MOTHER", "handshape": "HS_5_OPEN", "loc": [0.00, 0.08, 0.20], "mov": "CHIN_TAP"},
    {"lemma": "VISIT", "handshape": "HS_V_ALT", "start": [0.35, 0.45, 0.05], "end": [0.00, 0.00, 0.00]},
    {"lemma": "CAKEPOP", "oov_fallback": true, "procedural_tokens": {
      "idx": "CURVED", "mid": "CLOSED", "rng": "CLOSED", "pnk": "EXTENDED_SPREAD_R", "thm": "THM_PINCH_INDEX"}}
  ]
}
```

### 4.2.2 Stage 2: Modular Sub-Lexical Procedural Scaffolding & Biomechanical Muscle Engine

Stage 2 compiles the 4D JSON stream into a deterministic 52-joint SMPL-X skeletal scaffold [39]  $\hat{\mathbf{P}} \in \mathbb{R}^{T \times 52 \times 3}$  via analytical closed-form IK in  $\mathcal{O}(1)$  time, guaranteeing anatomical feasibility before neural processing:

1. **15-Joint, 20-DOF Anatomical Hand Model & Musculature:** Fingers are modeled as authentic 3-phalanx kinematic chains across 15 articulated phalangeal joints: 4 fingers with Metacarpophalangeal (MCP: 2-DOF flex  $[-15^\circ, 90^\circ]$ , abd  $[-20^\circ, 20^\circ]$ ), Proximal Interphalangeal (PIP: 1-DOF flex  $[0^\circ, 100^\circ]$ ), and Distal Interphalangeal (DIP: 1-DOF flex  $[0^\circ, 80^\circ]$ ) joints ( $4 \times 4 = 16$  DOFs); and the thumb articulating across Trapeziometacarpal/CMC (2-DOF flex/opposition), MCP (1-DOF), and Interphalangeal/IP (1-DOF) joints (4 DOFs), providing 20 active mechanical degrees of freedom alongside modeled thenar and hypothenar muscular pads.
2. **Articular Range-of-Motion (ROM) Limits:** Human motor control naturally minimizes joint strain. To guarantee anatomical feasibility, joint angles are penalized near their physiological limits, smoothly regularizing motion along the gradient of minimal strain and preventing awkward joint hyperextensions during trajectory synthesis. Dynamic trajectory smoothness and minimum-jerk optimization ( $\int \|\ddot{\mathbf{x}}\|^2 dt$ ) [20] are governed in continuous time by the residual diffusion score network.

3. **Multi-Digit Coordination & Soft-Body Collision Repulsion:** When distance between phalangeal segments falls below combined radii ( $r_A + r_B$ ), an elastic repulsion vector prevents self-intersection. Dynamic priority scheduling orders digit flexion in folded signs (e.g., in M, N, T, pinky and ring curl before the thumb crosses and index folds over).
4. **Analytical Arm Kinematics with Humeral Swivel Orbit:** Given shoulder root  $\mathbf{x}_{\text{shoulder}}$  and wrist target  $\mathbf{x}_{\text{wrist}}(t)$ , the shoulder anchor is first modulated by a 2-DOF clavicle articulation model ( $15^\circ$ – $20^\circ$  elevation and protraction) to prevent reach singularities at cranial minimal-pair targets (e.g., FATHER at the forehead). Elbow flexion is resolved via analytical 2-bone inverse kinematics in closed form, while humeral swivel redundancy parameterizes elbow position along a circular orbit orthogonal to the shoulder-wrist axis, regularized toward biological resting planes.
5. **Facial Blendshape Binding ( $F_{\text{NMS}}$ ):** Non-manual clause modifiers are mapped into 50 FLAME blendshapes [28] synchronized with the manual stroke phase.

#### 4.2.3 Stage 3: Manifold-Constrained Kinematic Residual Diffusion

Pure procedural animation is structurally correct but mechanically stiff. Stage 3 employs continuous diffusion solely to synthesize biological momentum, micro-coarticulation, and natural fluidity atop the rigid scaffold:

$$\mathbf{X}_{\text{final}} = \Pi_{\mathcal{M}}(\hat{\mathbf{P}} + \Delta\mathbf{X}_\theta) \quad (1)$$

where  $\Delta\mathbf{X}_\theta$  is the predicted residual micro-displacement. Grounded in **Dr. Farshad’s methodologies in topologically constrained generative models** (SCOPE [17], SceneGenie [15]), we enforce Riemannian manifold projection ( $\Pi_{\mathcal{M}}$ ) via iterative holonomic constraint projection using the SHAKE/RATTLE molecular dynamics algorithm and Neural SHAKE [2, 10, 40] (converging in  $\leq 3$ – $5$  iterations) adapted to skeletal kinematic trees:

1. **Bone-Length Invariance ( $\mathcal{L}_{\text{graph}}$ ):** Enforces exact distance constraints ( $\|\mathbf{x}_i(t) - \mathbf{x}_j(t)\|_2 = \ell_{ij}$ ) across all 52 skeletal connections and 15 phalangeal segments, permanently eliminating bone stretching.
2. **Character-Adaptive Contact Tolerance ( $r_{\text{tol}}$ ):** Enforces tight millimetric bounds ( $r_{\text{tol}} \leq 5$  mm) at critical contact landmarks (e.g., chin vs. forehead for MOTHER vs. FATHER), while permitting natural compliance (20–30 mm) during unconstrained movement in neutral space.
3. **Joint Limit Compliance:** Ensures neural residual updates respect the physiological range-of-motion limits established by the kinematic scaffold.

The denoiser is pretrained on **Motion-X++** (19.5M frames) [53] (regularized by  $\Pi_{\mathcal{M}}$ ), then fine-tuned on How2Sign’s 3D Panoptic subset ( $\sim 3$ h mocap) [12].

#### 4.2.4 Stage 4: Stylistic Prosody & Universal Cross-Lingual Transfer

A style vector  $\mathbf{s} \in \mathbb{R}^{512}$  modulates the denoiser via Adaptive Layer Normalization (AdaLN) to synthesize distinct signer personas, signing tempos, and emotional prosody. Swapping sub-lexical dictionary tables enables zero-shot cross-lingual transfer from ASL to Finnish Sign Language (SVK) and other sign languages with a detailed enough sub-lexical dictionary.

### 4.3 The Backward Pass: Sign-to-Text Language Translation (SLT)

The Backward Pass perceives and translates continuous signing video back into spoken text through three interpretable stages:

#### 4.3.1 Stage 1: Multi-Channel Perceptual Front-End & Domain-Optimized Tracking

Operating over dedicated signers facing the camera, the perceptual front-end recovers 3D anatomical articulators:

1. **20-DOF Hand Mesh Reconstruction:** High-dexterity tracking via HaMeR [54] extracts rotational joint angles across all 20 active mechanical degrees of freedom (15 articulated joints per hand).
2. **Facial Action Units & Gaze Vector:** Tracks 50 FLAME blendshapes [28] (eyebrows, mouth morphemes) and the 3D ocular gaze unit vector  $\mathbf{g}_{\text{gaze}}(t) \in \mathbb{R}^3$ .
3. **Body-Centric Sternum Normalization Frame:** Skeletal keypoints  $\mathbf{x}_{\text{world}} \in \mathbb{R}^3$  are normalized to an invariant coordinate frame centered at the signer’s sternum ( $[0, 0, 0]$ ) and scaled by biacromial shoulder width ( $W_{\text{shoulder}}$ ), eliminating perspective and scale variances.

#### 4.3.2 Stage 2: 3D Spatial Loci Raycasting & Central Memory Bank ( $M_t$ )

Stage 2 tracks spatial interactions and maintains discourse memory across turns:

- **Palm & Gaze Aiming Raycasting:** Continuous 3D aiming rays are cast from the normalized palm orientation vector ( $\vec{v}_{\text{aim}} = \mathbf{R}_{\text{wrist}} \cdot \mathbf{u}_{\text{point}}$ ) and ocular gaze vector  $\mathbf{g}_{\text{gaze}}(t)$ , intersecting active 3D spatial loci in the interaction volume.
- **Decoupled Central Memory Bank ( $M_t$ ):** Active referents ( $K \leq 8$  entities) are stored in an external stateful memory bank:

$$M_t = \{\mathbf{m}_k = \langle \mathbf{x}_k, \mathbf{e}_k, \alpha_k, \tau_k \rangle\}_{k=1}^K \quad (2)$$

where  $\mathbf{x}_k \in \mathbb{R}^3$  is the physical coordinate anchor,  $\mathbf{e}_k$  is the entity embedding,  $\alpha_k$  is the salience weight, and  $\tau_k$  is the last interaction timestamp. Decoupling  $M_t$  into an  $\mathcal{O}(1)$  external register allows entities to persist across long dialogues without inflating the LLM context or causing spatial state collapse when sliding temporal windows advance. Crucially, while articulatory tracking is normalized to the sternum for scale invariance, spatial referents  $\mathbf{x}_k$  in  $M_t$  are parameterized in an egocentric horizontal ground-plane frame decoupled from torso yaw, preventing discourse anchors from erroneously rotating when a signer executes a physical role-shift ( $15^\circ$ – $30^\circ$  torso turn).

- **Deictic Frame-of-Reference Inversion ( $\mathbf{R}_y(\pi)$ ):** In face-to-face dialogue, the coordinate frames of signer and avatar face each other along the line of sight. To preserve biological body and hand chirality while aligning spatial reference loci during turn-taking, we apply a proper rigid  $SO(3)$  deictic rotation with conversational offset  $\mathbf{t}_{\text{conv}}$ :

$$\mathbf{x}_{\text{avatar}} = \mathbf{R}_y(\pi) \cdot \mathbf{x}_{\text{signer}} + \mathbf{t}_{\text{conv}}, \quad \mathbf{R}_y(\pi) = \text{diag}(-1, 1, -1) \quad (3)$$

Crucially, because  $\det(\mathbf{R}_y(\pi)) = (-1)(1)(-1) = +1 \in SO(3)$ , this transformation preserves physical chirality (a right hand remains an authentic right hand), while correctly mapping the signer’s right hemisphere (contralateral to the addressee’s left visual field) to the corresponding shared deictic locus.

- **Temporal Smoothing & Thematic  $\theta$ -Role Matching:** To mitigate high-frequency monocular tracking noise without sacrificing precision, aiming rays are temporally integrated across a 5–10 frame window ( $\sim 160$  ms). Intersection with active referents in  $M_t$  is resolved via an adaptive Gaussian tolerance kernel  $\mathcal{N}(\mathbf{x}_k, \Sigma_{k,t})$  updated dynamically by an Extended Kalman Filter (EKF). Furthermore, phonological contrast in natural sign discourse ensures that distinct entities inhabit well-separated canonical signing sectors (contralateral, center, ipsilateral) with angular separation  $> 30^\circ$ , reliably exceeding the monocular tracking uncertainty cone. Directional verbs are resolved across three phonological classes [32]: Path verbs (GIVE) via trajectory start/end ( $\mathbf{p}_{\text{start}} \rightarrow \mathbf{p}_{\text{end}}$ ); Backward verbs (INVITE) via lexical mapping inverting Goal to Source; and Orientation verbs (LOOK-AT) via  $\vec{v}_{\text{aim}}$ .

#### 4.3.3 Stage 3: LLM Translation, Memory Management & Scene-Graph Grounding

Stage 3 decodes fluent text while governing memory and grounding gestures in the physical environment:

- **Memory Bank Updates & Eviction:** The foundation model maintains  $M_t$  dynamically: registering new loci upon introduction, refreshing salience weights upon re-reference, and evicting completed topics via soft exponential decay in  $\mathcal{O}(1)$  time.
- **Environmental Grounding via Semantic Scene Graphs (*SceneGenie* [15]):** When a signer points to a real-world object (e.g., a coffee cup, door, or medical tool) or uses an ambiguous polysemic depicting classifier (CL:C), skeletal tracking cannot disambiguate the referent. We structure the room environment into a 3D semantic scene graph  $\mathcal{G}_{\text{env}} = (\mathcal{V}_{\text{obj}}, \mathcal{E}_{\text{rel}})$ . Intersecting the pointing ray  $\vec{v}_{\text{aim}}$  with object bounding volumes allows the LLM to instantly resolve real-world referents without heavy raw video compute.

#### 4.4 The BiSign-LLM: Unified 3D Spatial-Lexical Foundation Modeling

Bidirectional synergy between continuous physical kinematics and symbolic linguistics is unified through the **Bidirectional Spatial Sign Large Model (BiSign-LLM)**. Rather than treating sign processing as disconnected translation and production pipelines, we formulate continuous sign language as a **multimodal language modeling task over a unified token vocabulary**.

**Adapting Pre-Trained Foundation Models vs. Training from Scratch:** A critical architectural decision is whether to pre-train a Transformer from scratch or adapt an existing frontier model. Training an 8B-parameter Transformer from scratch exclusively on sign video tokens is mathematically sub-optimal: even aggressive web-scraping yields billions of sign tokens, whereas learning general world knowledge, commonsense reasoning, and multilingual syntax from scratch demands  $\geq 10$ – $15$  trillion tokens. A model trained from scratch would lack basic real-world comprehension (e.g., medical concepts, civic

procedures, spatial commonsense). We therefore adapt an existing open-weight frontier foundation model—**Llama 3.1 8B** or **Mistral NeMo 12B**—inheriting its deep factual knowledge and multilingual reasoning while teaching it the physical modality of sign language through vocabulary expansion, augmenting the base text tokenizer with discrete sub-lexical phonological tokens (handshapes, finger curls, orientation vectors) and spatial coordinate bins discretizing the interaction volume  $[-1, 1]^3$  [6, 11]. The new token embeddings  $\mathbf{z} \in \mathbb{R}^{d_{\text{model}}}$  are initialized via a linear projection from their geometric definitions, ensuring warm optimization.

#### Web-Scale Video Harvesting & 3-Stage Optimization on EuroHPC LUMI:

1. **Stage 1: Web-Scale Token Harvesting & Continual Pre-Training:** Module 1 (Open-Domain Perceptual Tokenizer) crawls unlabelled signing videos at scale from YouTube, Instagram Reels, and broadcast media. By tracking 20-DOF hands, 50 FLAME blendshapes, and gaze vectors, continuous video is automatically converted into millions of discrete token sequences  $S_{1:T} = [s_1, \dots, s_T]$ . We perform continual self-supervised pre-training across these token sequences on CSC’s **EuroHPC LUMI supercomputer** (AMD MI250X nodes, PyTorch FSDP + ROCm FlashAttention). Through this stage, the LLM masters the native transition probabilities, spatial co-articulation, and syntactic rhythm of natural sign language without requiring manual sentence-level text annotations.
2. **Stage 2: Bidirectional Multi-Task Instruction Tuning:** The continually pre-trained foundation model is fine-tuned on aligned multimodal parallel datasets (How2Sign [12], PHOENIX-2014T, ASL-LEX pairs) formatted into unified prompt templates:
  - ▶ **Sign-to-Text (SLT):** [SIGN]  $s_1 \dots s_k$  [TRANSLATE]  $\rightarrow t_1 \dots t_m$  (English/Finnish text).
  - ▶ **Text-to-Sign (SLP):** [TEXT]  $t_1 \dots t_m$  [SIGN]  $\rightarrow s_1 \dots s_k$  (1–2 Hz discrete sign token blueprint).
  - ▶ **Conversational Sign-to-Sign:** [PROMPT\_SIGN]  $s_1 \dots s_k \rightarrow s'_1 \dots s'_n$  (Direct native sign dialogue).
3. **Stage 3: Community Preference Optimization (DPO):** To eliminate unnatural transitions and align avatar prosody with cultural norms, the model undergoes Direct Preference Optimization (DPO) grounded in human feedback from native Deaf consultants at *Kuurojen Liitto ry*.

#### 4.5 Empirical De-risking: Kinematic Synthesizer Compilation & Phonological Laws

As established in the Day-1 prototype highlighted on Page 1, the interactive **Kinematic Synthesizer (WebGL Demo)** confirms that sub-lexical symbolic compilation into continuous 3D articulation is computationally efficient and anatomically compliant. Running at 60 Hz in client-side WebGL, the engine compiles discrete phonological descriptions into deterministic 3D hand poses in  $\mathcal{O}(1)$  time ( $< 0.5$  ms) under anatomical joint limits and tendon spring dynamics.

Crucially, the prototype operationalizes **Battison’s Classical Phonological Laws (1978)** [4] across a two-handed lexicon:

1. **Symmetry Constraint:** For symmetrical signs (BOOK, MEET, PLAY, FAMILY, SAME, PEACE), both hands share identical anatomical handshapes and execute synchronous, bilaterally mirrored trajectories.
2. **Dominance Constraint:** For asymmetrical signs (WORK, HELP, STOP, NAME, COFFEE, LOVE), the non-dominant Left hand remains a stationary spatial base anchor while the dominant Right hand articulates atop it.
3. **Live Biomechanical Joint Strain Monitoring:** The engine evaluates joint displacement relative to resting comfort angles across both hands, providing immediate visual telemetry of articular limits and ensuring poses remain within natural physiological envelopes without hyperextension.

Furthermore, for novel out-of-vocabulary terms (‘CakePop’), the foundation model operates strictly at the discrete phonological level—emitting sub-lexical tokens ( $\text{Tokens}(\text{CakePop}) = \{\text{idx} : \text{CURVED}, \text{pnk} : \text{SPREAD\_R}, \text{thm} : \text{PINCH\_INDEX}\}$ ) which our kinematic engine compiles deterministically into anatomically valid 3D postures. This confirms that the geometric foundation, 3D graphics rendering, and inverse kinematics pipelines are already operational and de-risked.

## 5 Methodological Synergy with Target Lab at Aalto University & ELLIS Finland

Continuous sign language processing represents a high-impact application domain for **Dr. Azade Farshad’s** research group at **Aalto University / ELLIS Institute Finland**. As established in Section 4, our bidirectional neuro-symbolic framework directly integrates and extends her group’s pioneering methodologies in geometric deep learning and generative modeling:

1. **Scene-Graph Generative Conditioning (Extending SceneGenie [15]):** In *SceneGenie*, Dr. Farshad

demonstrated that conditioning diffusion models on structured semantic scene graphs significantly improves visual coherence. Sign discourse syntax operates precisely as a 3D spatial scene graph  $\mathcal{G} = (\mathcal{V}_{\text{entities}}, \mathcal{E}_{\text{relations}})$ , where nodes encode coordinate loci ( $L_1, L_{3a}, L_{3b}$ ) and edges encode directional agreement predicates. Encoded via a Relational Graph Transformer (RGT), these embeddings condition the residual diffusion denoiser, eliminating minimal-pair drift. In reverse, scene graphs ground physical room objects to disambiguate pointing gestures.

2. **Topologically Constrained Generative Modeling (Extending *SCOPE* [17] & *HieraSurg* [16]):** In surgical video generation and anatomical segmentation, Dr. Farshad established that biological structures require explicit graph-based continuity regularizers. High-DOF articulated hands and arms exhibit identical topological vulnerabilities during neural diffusion. Adapting these graph constraints guarantees bone-length invariance ( $\mathcal{L}_{\text{graph}}$ ) and physiological strain compliance, preventing joint dislocations.
3. **ELLIS Priority “AI for Health & Society”:** According to the WHO, health encompasses complete physical, mental, and social well-being. Chronic communication isolation represents a severe social determinant of health. This bidirectional system acts as an **Assistive Communication Digital Twin**, dismantling systemic exclusion across healthcare, public services, and daily life.

### 5.1 Open Research Questions for Collaborative Investigation

While our framework establishes clear mathematical interfaces between physical kinematics and multimodal reasoning, two fundamental scientific questions represent genuine unknowns that I seek to investigate in close collaboration with **Dr. Azade Farshad**:

1. **Riemannian Manifold Projections: Rigid Invariance vs. Soft-Tissue Non-Rigidity:** In *SCOPE* [17], topological graph constraints enforce geometric continuity across rigid anatomical structures. Articulated hands, however, exhibit a dual mechanical nature: rigid skeletal phalanges coupled with continuous, non-rigid soft-tissue contact (fleshy pulp deformation, thenar pad compression during closed grips, and variable inter-digital compliance). A key theoretical question is how to generalize the holonomic projection operator  $\Pi_{\mathcal{M}}$  from rigid pairwise distance matrices to unilateral inequality constraints and elastodynamic contact manifolds without inducing high-frequency gradient chatter or inflating diffusion sampling latency during real-time generation.
2. **Dynamic Scene-Graph Diffusion for Non-Stationary Discourse:** While *SceneGenie* [15] formulates generative diffusion conditioned on static or piecewise-constant 3D scene graphs, conversational sign language demands a *continuous, dynamically mutating topology*  $\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t)$ . Discourse referents are instantiated, spatialized, and evicted at sub-second conversational rates, causing the conditioning graph’s adjacency matrix to change asynchronously. Formulating a spatio-temporal graph diffusion architecture that guarantees trajectory smoothness and semantic stability across discrete topological mutations is an open methodological challenge directly aligned with Dr. Farshad’s geometric deep learning research.

## 6 Multi-Tier Evaluation Framework

To benchmark generative biomechanical accuracy (SLP) and translation fidelity (SLT), we establish the multi-tier framework summarized in Table 2. Crucially, existing continuous sign datasets (How2Sign [12], PHOENIX-2014T) contain text transcripts and keypoints, but lack 3D spatial referent ground truth (no labeled 3D  $R$ -loci or co-reference chains). Rather than conditioning doctoral milestones on massive manual re-annotation, our primary validation uses **Task-Grounded Downstream Evaluation** via **Argument Translation F1** (subject-object pronoun precision/recall in English/Finnish) and **Agreement-Gated BLEU-4**, isolating sentences strictly requiring spatial agreement. As a secondary direct check, if feasible within community partnership resources, we pilot a targeted **Spatial Agreement Benchmark (SAB-100)**, annotating 100 multi-turn clips with Deaf linguists from *Kuurojen Liitto ry* (scalable up to 500 clips if resources permit).

### 6.1 Downstream Task-Grounded Validation Strategy

Standard natural language processing benchmarks frequently evaluate sign translation using n-gram metrics such as BLEU-4. In sign linguistics, however, BLEU-4 exhibits a severe blind spot: a translation may achieve a high lexical score by correctly translating vocabulary items while completely confusing subject and object pronouns due to misattributed directional verbs [30, 38]. To provide rigorous

| Dimension                     | Evaluation Metric & Protocol                                                                                   | Validation Objective & Target                                                   | Comparative Baselines                                               |
|-------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Landmark Precision (SLP)      | <b>End-Effector Error (EFE<sub>mm</sub>)</b> : Mean Euclidean distance on fingertip/cranial contact landmarks. | Significant error reduction; target calibrated via initial pilot runs.          | Discrete diffusion (G2P-DDM [50]), Neural Sign Actors [3]           |
| Kinematic Realism (SLP)       | <b>Fréchet Video Distance (FVD)</b> and Mean Kinematic Jerk ( $\int \ \ddot{x}\ ^2 dt$ ).                      | Minimize ballistic jitter; comply with biological minimum-jerk profiles.        | Unconstrained motion diffusion (MDM [47]), PhysDiff [52]            |
| Minimal-Pair Fidelity (SLP)   | Spatio-temporal classifier accuracy distinguishing minimal pairs (MOTHER vs. FATHER).                          | Eliminate semantic inversion; preserve contact-location boundaries.             | Unconstrained skeletal diffusion models without manifold projection |
| Spatial Locus Grounding (SLT) | <b>Discourse Tracking Accuracy</b> : Spatial Locus Alignment (SLA) across conversational turns.                | Zero spatial-state collapse across multi-turn referents; pilot on SAB-100.      | Sliding-window SLT transformers [38, 51]                            |
| Agreement Resolution (SLT)    | <b>Argument Translation F1</b> : Precision/recall on pronoun and thematic $\theta$ -role binding.              | Robust resolution across directional, backward, and orientation verbs.          | Continuous sign translation models (CorrNet+ [24], STMC [51])       |
| Continuous Translation (SLT)  | <b>BLEU-4 &amp; Word Error Rate (WER)</b> on public benchmarks (How2Sign, PHOENIX-2014T).                      | Target threshold calibrated against published SOTA following pilot.             | Standard multimodal sequence-to-sequence translation architectures  |
| Inference Latency (System)    | End-to-end streaming throughput (fps) and wall-clock latency (ms) on single GPU.                               | Real-time interactive execution ( $\geq 60$ fps, $< 100$ ms streaming latency). | Monolithic autoregressive vision-language foundation models         |
| Deaf Community Quality        | Structured Likert Evaluation (1–5 scale) with certified Deaf consultants from <i>Kuurojen Liitto ry</i> .      | Qualitative threshold on spatial clarity and naturalness set with Deaf panel.   | Procedural animation avatars [31] and unconstrained synthesizers    |

Table 2: Multi-Tier Evaluation Matrix balancing biomechanical precision, spatial discourse tracking, and community validation.

scientific evaluation without depending on massive manual spatial re-annotation, our methodology emphasizes **Argument Translation F1**. We construct an evaluation subset isolating transitive sentences with third-person referents, measuring whether grammatical subject and object assignments in translated English or Finnish match the ground truth. This task-grounded protocol directly assesses whether the perceptual tokenizer and causal memory bank correctly resolve 3D spatial loci without requiring manual 3D coordinate annotations for training.

## 7 4-Year Doctoral Research Work Plan: Parallel Tracks & Staggered Milestones

Adhering to Aalto University’s article-based doctoral dissertation guidelines (structured around a cohesive series of peer-reviewed publications in top international venues), the research progresses across two foundational tracks in Years 1–2 that provide the physical actuator (Track A) and perceptual tokenizer (Track B), converging in Year 3 into the unified foundation model (**Track C**). The doctoral plan is explicitly focused on **Tier 1 (Core Dissertation Milestone)** as the non-negotiable scientific requirement:

- ▶ **Years 1–2: Foundational Actuator & Perceptual Tokenizer Modules (Tier 1 Core)**
  - ▶ *WP1 (M1–M16, Track A: Biomechanical Actuator & Kinematic Scaffolding)*: Implement Kinematic Synthesizer kinematics (30 articulated joints) with Battison phonological laws and closed-form 2-bone arm IK. Deploy Manifold-Projected Residual Diffusion ( $r_{\text{tol}} \leq 5$  mm,  $\mathcal{L}_{\text{graph}}$ ). **Paper 1 (M17, Target: ECCV 2028)**: *Anatomically Constrained Residual Diffusion for Articulated Sign Kinematics*.
  - ▶ *WP2 (M1–M16, Track B: Web-Scale Perceptual Tokenizer & Causal Memory)*: Crawl open-domain signing videos. Deploy HaMeR 20-DOF hand tracking, 50 FLAME blendshapes, and sternum normalization. Build decoupled  $\mathcal{O}(1)$  Causal Spatial Memory Bank ( $M_t$ ) with deictic rotation  $R_y(\pi)$  and *SceneGenie* scene-graph grounding.
- ▶ **Year 3: Foundation Model Adaptation & Pre-Training on EuroHPC LUMI (Tier 1 Core)**
  - ▶ *WP3 (M17–M26, Track B/C: Dual-Rate Streaming & Vocabulary Expansion)*: Expand open-weight LLM tokenizer (Llama 3.1 8B / Mistral NeMo 12B) with joint spatial-lexical vocabulary  $\mathcal{V}_{\text{extended}} = \mathcal{V}_{\text{base}} \cup \mathcal{V}_{\text{sign}}$ . Integrate 60 Hz kinematics with 1–2 Hz macro-tokens.
  - ▶ *WP4 (M24–M32, Track C: BiSign-LLM Continual Pre-Training on EuroHPC LUMI)*: Continually pre-train extended foundation LLM on EuroHPC LUMI (AMD MI250X nodes, PyTorch FSDP + ROCm) across harvested sign tokens. Multi-task instruction tuning (Sign  $\leftrightarrow$  Text). **Paper 2 (M33, Target: EMNLP 2029)**: *BiSign-LLM: Adapting Open-Weight Foundation Models for Continuous Sign Language AI*.
  - ▶ *WP5 (M31–M36, Track C: Downstream Validation & Core Synthesis)*: Task-grounded downstream evaluation (Argument F1, Agreement-Gated BLEU, phonological minimal pairs). **Paper 3 (M36, Target: ICLR 2030)**: *Multimodal Spatial Foundation Modeling for Open-Domain Assistive Communi-*

cation.

► **Year 4: Pre-Examination, Community Synthesis & Statutory Defense**

- *WP6 (M37–M41, Oct 2029–Feb 2030)*: Pilot evaluation with certified Deaf consultants and preparation of open-source foundation release.
- *Statutory Pre-Examination & Defense (M42–M48, Mar–Sept 2030)*: Cumulative dissertation compilation, formal pre-examination (*esitarkastus*) by external opponents, and public defense (*väitöstilaisuus*, Month 48) at Aalto University.

**Prospective Collaborative Extensions (Tiers 2–3: Beyond Core Dissertation Scope)**: To guarantee doctoral completion, core requirements (Tier 1) are strictly confined to the ASL foundation pipeline, manifold residual diffusion, causal spatial memory bank, and automated task-grounded evaluation. Secondary objectives—including zero-shot cross-lingual transfer to Finnish Sign Language (**SVK**), multi-session community MOS user studies, and production-grade conversational avatar deployment—represent **prospective collaborative upside** pursued with master’s students and *Kuurojen Liitto ry* without endangering the core doctoral defense timeline.

## 8 Computational Feasibility, Ethical Governance & Risk Mitigation

### 8.1 Scope Calibration: Why This Scope, and Not More

A rigorous doctoral proposal must define what it explicitly excludes to prevent shallow execution. We confine this research strictly to the foundational representation problem—uniting continuous physical biomechanics with discrete multimodal foundation models via manifold-projected kinematics and causal spatial memory—bounding the scope across three dimensions:

1. **No Photorealistic Neural Rendering (Avatar Mesh, Not NeRFs)**: We focus on 3D skeletal joint trajectories, mesh blendshapes, and WebGL kinematics. Photorealistic pixel-level neural rendering (4K Gaussian Splatting or neural textures) is computationally decoupled; the kinematic scaffold drives any third-party engine.
2. **No Primary Linguistic Corpus Collection from Scratch**: We leverage existing high-quality datasets (How2Sign, PHOENIX-2014T, Motion-X++) and web-scale unlabelled video harvesting for pretraining, avoiding de-novo national corpus collection.
3. **Single-Signer Articulation Before Multi-Party Interaction**: Core milestones address single-signer production and translation. Conversational multi-party turn-taking is evaluated in a pilot setting via deictic inversion ( $\mathbf{R}_y(\pi)$ ), treating full multi-agent spatial coordination as post-doctoral research.

**High-Performance Computing Strategy (EuroHPC LUMI)**: Kinematic pretraining on Motion-X++ (19.5M video-estimated frames regularized via  $\Pi_{\mathcal{M}}$ ) [53] and BiSign-LLM continual pre-training leverage Aalto’s direct access to the **EuroHPC LUMI consortium** (CSC, Kajaani, Finland). LUMI-G nodes feature **AMD Instinct MI250X GPUs** (each node delivering 8 Graphics Compute Dies with 512 GB HBM2e memory). The software stack utilizes native **AMD ROCm 6.x / HIP**, Composable Kernel, FlashAttention-2 ROCm port, and PyTorch Fully Sharded Data Parallel (FSDP) with mixed-precision (bf16) and activation checkpointing. Training datasets are hosted on LUMI’s parallel **Lustre filesystem** (/scratch) packaged into indexed **WebDataset** tar shards, ensuring high sustained I/O throughput during multi-node distributed training.

**Ethical Governance & Deaf Community Participatory Co-Design**: Formal ethical review is conducted by the **Aalto University Research Ethics Committee** prior to participant evaluations. In alignment with **Finland’s Sign Language Act** (*Viittomakielilaki 359/2015*), research involving sign language must respect linguistic autonomy and community cultural sovereignty. In partnership with *Kuurojen Liitto ry*, an ongoing **Deaf Advisory Board** provides continuous feedback on blendshapes, cultural nuance, and linguistic naturalism from Month 1.

**3-Tier Contingency Fallback Hierarchy**: To guarantee timely doctoral completion adhering to Aalto’s article-based dissertation standards: (1) *Tier 1 (Core Non-Negotiable)*: Full ASL bidirectional procedural-diffusion pipeline, causal memory bank, mathematical loss formulations, and automated quantitative benchmarking (Years 1–3). (2) *Tier 2 (Cross-Lingual Flexibility)*: If SVK extraction encounters friction, transfer focuses on calibrated ASL agreement benchmarks before scaling to open-domain SVK continuous discourse. (3) *Tier 3 (User Study Scalability)*: If participant recruitment bottlenecks arise, evaluation transitions flexibly to a panel of 5 certified Deaf interpreters using standardized Likert rubrics.

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